

Project Documentation: Infosys Stock AI

1. Introduction

Infosys Stock AI is a comprehensive financial analytics platform designed to bridge the gap between raw market data and actionable insights. By integrating a robust **FastAPI** backend with advanced Machine Learning architectures—including **Deep Learning (LSTM)**, **Ensemble Methods (XGBoost, Random Forest)**, and **Statistical Models (ARIMA, Prophet)** - the system provides users with real-time market tracking and multi-model price forecasting.

The platform features an interactive dashboard for visualizing historical data, an AI-powered chatbot for natural language market queries, and a secure authentication system, making complex financial analysis accessible to both novice and experienced users.

2. System Overview

The system operates as a full-stack web application that automates the entire data pipeline: from extraction and preprocessing to model inference and visualization.

Core Capabilities:

- **Dynamic Data Retrieval:** Instantly fetches live and historical data for global stocks (NSE, BSE, NYSE, NASDAQ) and cryptocurrencies using the **yfinance** engine.
- **Multi-Model Forecasting:** Allows users to simulate future prices by selecting specific target dates and choosing from six distinct algorithms to compare performance.
- **Intelligent Chat Assistance:** A built-in chatbot parses natural language to provide instant answers on price trends, 52-week ranges, and simple linear predictions.
- **Visual Analytics:** Interactive line and candlestick charts with adjustable timeframes (1M, 1Y, 5Y) powered by ApexCharts.

3. Technology Stack

3.1 Frontend

- **Core:** HTML5, CSS3, JavaScript (ES6+).
- **Visualization:** **ApexCharts.js** for dynamic, responsive financial charting.
- **Architecture:** Component-based UI with modular CSS (index.css, forecast.css, chatbot.css).

3.2 Backend

- **Framework:** **FastAPI** (Python) for high-performance, asynchronous API delivery.
- **Server:** Uvicorn (ASGI server).
- **Data Processing:** Pandas, NumPy, yfinance.
- **Security:** Argon2 hashing via passlib for secure user authentication.

3.3 Machine Learning Models

The system incorporates a "Model Registry" architecture to dynamically load and execute the following models:

1. **LSTM (Long Short-Term Memory):** Deep learning model optimized for time-series sequence prediction.
2. **Prophet:** Additive regression model best for seasonality and holiday effects.
3. **XGBoost:** Gradient boosting framework for efficient, structured data regression.
4. **ARIMA:** Statistical analysis for non-seasonal stationarity.
5. **Random Forest:** Ensemble learning method for regression.
6. **Linear Regression:** Baseline model for trend estimation.

4. System Architecture

The application follows a modular Microservices-like architecture:

1. **Authentication Module:** Handles user signup and login. Passwords are hashed using Argon2 before storage in users.json.
2. **Market Data Module:** Serves as the interface to the Yahoo Finance API. It handles requests for specific tickers, normalizes OHLC (Open, High, Low, Close) data, and manages time-period filtering.
3. **Prediction Engine:**
 - Receives a ticker and target date from the frontend.
 - Fetches the required training data (last 2 years).
 - Routes the data to the user-selected model function (e.g., predict_lstm).
 - Returns forecast values and accuracy metrics (RMSE).
4. **Chatbot Engine:**
 - Uses Regex to extract tickers (e.g., "INFY.NS") and dates from user messages.
 - Computes technical indicators like the **50-day Simple Moving Average (SMA)** to determine Bullish/Bearish trends.
 - Runs lightweight Linear Regression for on-the-fly predictions.

5. Functionality

5.1 Dashboard (Market Watch)

- **Search:** Users can search for any stock symbol (e.g., AAPL, TCS.NS).
- **Metrics:** Displays real-time Open, High, Low, Close, and Volume data.
- **Quick View:** Sidebar containing pre-set buttons for popular Indian and US stocks.

5.2 Forecast Studio

- **Custom Inputs:** Users select a stock ticker and a specific future date.
- **Model Selection:** Users can toggle specific models to run (e.g., "Compare LSTM vs. XGBoost").
- **Visualization:** Results are plotted alongside historical data to visualize the "prediction cone".

5.3 AI Chatbot

The chatbot supports conversational queries such as:

- *"What is the price of Wipro?"*
- *"Is Reliance Bullish or Bearish?"* (analyzes trend vs 50-day SMA).
- *"Predict TCS for 2025-12-01"*
- *"Show me the 52 week high of Google"*

6. System Output

- **Operational Output:** JSON-formatted API responses containing stock metadata, historical price arrays, and forecast objects.
- **Visual Output:**
 - **Candlestick Charts:** For detailed technical analysis.
 - **Area/Line Charts:** For macro trend visualization.
 - **Prediction Graphs:** Multi-line overlays showing how different models diverge in their forecasts.

7. Future Enhancements

- **Portfolio Management:** Allow users to save stocks to a persistent "Watchlist" database.
- **Sentiment Analysis:** Integrate news APIs to adjust predictions based on market sentiment scores.
- **Deployment:** Containerize the application using Docker for cloud deployment (AWS/Azure).
- **Advanced Metrics:** Display RSI, MACD, and Bollinger Bands on the main chart.

8. Conclusion

Infosys Stock AI demonstrates the effective fusion of traditional web technologies with modern Data Science. By abstracting complex ML pipelines behind a user-friendly API and interface, it empowers users to make data-driven decisions without needing expertise in coding or statistics. The modular design ensures scalability, allowing for the easy addition of new models and features in the future.